

Home Studio: A Mixed Reality Staging Tool for Interior Design

Christian Vazquez
cvazquez2@wayfair.com
Wayfair
Boston, US

Nicole Tan
ntan@wayfair.com
Wayfair
Boston, US

Shrenik Sadalgi
ssadalgi@wayfair.com
Wayfair
Boston, US



Figure 1: Home Studio allows interior design services to stage any publicly available 3D Matterport scene with Wayfair products and produce photorealistic renderings in minutes.

ABSTRACT

Home design services employ virtual real estate staging to visually convey proposals to prospective customers, instilling confidence in property owners on how spaces will look once they are furnished. However, creating room-scale visualizations requires days of expert labor to composite products on images provided by the customer. Advances in 3D capture now let users scan their spaces with a smartphone in a matter of minutes, enabling scale-accurate mixed reality experiences that can be leveraged to lower the skill bar and time required to produce visualizations of furniture in the user's context. We present Home Studio, a web-based tool that lets designers stage any Matterport scan and generate photorealistic renders of products in the user's context. Our tool enables a drag and drop experience that reconstructs the perspective view of the Matterport scene in a remote service, employing a rendering engine to produce a photorealistic composite.

CCS CONCEPTS

• **Applied computing** → Online shopping; • **Computing methodologies** → Rendering; Mixed / augmented reality.

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KEYWORDS

Mixed Reality; Online Retail; Rendering

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1 INTRODUCTION

Mixed reality experiences help users assess the fit or visual appeal of products within a particular context [4, 36], by creating true to life visualizations that instill confidence in customers on their purchase decisions and narrow the imagination gap required to envision bold, new projects [34]. In real estate, virtually staged homes are sold 87% faster than non-staged listings, making it an attractive incentive to realtors and prospective homeowners alike [17, 21]. Accelerated by the Covid-19 pandemic, more 3D scans of real estate are becoming readily available to the general public as a means to enable virtual tours to potential clients and comply with social distancing guidelines [35]. However, generating virtually staged content requires expert 3D labor to achieve high quality imagery that is scale-accurate.

Interior design services, such as those offered by Mody [22] and Roomy [31], provide customers designs of their spaces staged with manually curated products, using photorealistic renders or composites to sell the “look” of a proposal. Customers provide sufficient imagery of the space they want to see furnished and the service returns either furniture composed into those images or renders of

a recreation of the space with the target products. The manual effort and expertise required to produce photorealistic visualizations translates into increased turnaround times, ranging in the order of days to weeks [1, 2]. Moreover, the 3D skill necessary to achieve quality results is a non-starter for parties such as realtors, homeowners, and interior designers to service their own visualization needs without outsourcing the work.

Advances in 3D capture technology are making scans of real estate feasible for the average consumer. Companies like Matterport [19] and Occipital [23], have developed solutions that let customers accurately scan their homes with their smartphones. The inclusion of dedicated hardware (e.g. Lidar), in consumer devices lays the foundation to "home in the cloud" as a paradigm that enables context-rich experiences driven by a digital twin of the user's home. The availability of software development kits (SDK) that let developers integrate their content with these 3D captures, such as the Matterport Bundle SDK [20], afford the creation of mixed reality tools that empower consumers to generate meaningful visualizations from scale accurate 3D spaces.

In this work, we present Home Studio, a web-based tool that leverages Matterport's scanning and visualization capabilities (typically employed for virtual tours) to enable a mixed reality staging experience. Our tool decreases the 3D skill barrier required to create photorealistic visualizations of home goods in real estate, allowing designers, homeowners, and non-experts to bring their designs to life in minutes.

2 RELATED WORK

In the context of interior design, visualization tools target varying skill levels. For novice users, these range from simple overlays on floor plans [9, 30] to mixed reality applications for in-situ staging [4, 34, 36]. Expert-level tools like CAD provide more realistic and accurate visualizations of spaces [28, 33]. We present the contribution of our work on two fronts.

First, HomeStudio's simple drag-and-drop interface on top of the Matterport scene enables designers of any level to create accurate [18], room-scale visualizations quickly. Second, our rendering system allows for higher levels of photorealism in comparison to traditional manual and mixed reality design tools.

Photo-based applications allow users to capture an image and then overlay 3D models. Examples include Fournier [10] and Karsh's [15] systems as well as Adobe Dimension [3], which all utilize depth

estimation to recreate the scene layout prior to object insertion. With these tools, designers are limited to a narrow, set window for staging as the field of view is restricted to a single image. As our work is instead focused on larger, room-scale utility for interior designers, building upon Matterport scans allows for entire homes to be furnished. Designers are additionally empowered to holistically represent their work; staging not just the scene but the shot angles needed to give the customer the best representation of their design.

Similar to these photo-based overlay tools, mixed reality applications have been developed for in-situ staging, including Park et al's DreamHouse interior design system [24], RoomAlive [13], and SceneCtrl [37]. A subset of such tools are augmented reality applications, which allow consumers to clarify fit and style of furniture, using only their mobile devices to place products in their space [6]. Customers can easily visualize single products from multiple vantage points by walking around, providing more contextualization pre-purchase. Home Studio provides for similar interactivity by allowing users to freely navigate the 3D space, but does not constrain the designer to being physically present during furnishing. Additionally, mixed reality applications often leave photorealism to the wayside due to hardware constraints [7, 26]. Home Studio differs in our use of a proxy-based distributed rendering paradigm, substituting lower fidelity models with their higher fidelity counterparts during rendering. As novice users can do this with the click of a button, it empowers them to create photorealistic output without the need to understand lighting, shadowing, or even 3D rendering.

Much research has been done in creating photorealistic composites from these overlays, focusing mainly on determining the lighting of a given photograph and applying it to 3D models for visual coherence. Examples include DeepLight [14], Gardner et al's system for inferring indoor illumination from low dynamic range images [11], among many others [8, 12, 16, 27, 29]. Research has also been done in utilizing proxies as substitutes for renders, as seen in Pomi's system [25]. Our work differs in that our usage of Matterport as the base application layer provides the structural features of the scene without the need to reconstruct those properties from a photograph.

Design services, such as Modsy [22], create photorealistic imagery, but are dependent on manual artist labor to recreate structural scene properties from photographs. Rooomy [31] is similar, but instead utilizes Matterport scenes for staging, much like our tool. We differ on two fronts: first, our tool is designer-facing, acting as a one-stop shop for dragging and dropping furniture into any Matterport scene; design services instead focus on the collaboration between clients and design experts. Second, due to both constant interaction and manual artist labor, turnaround times for design services can be upwards of a week, whereas our tool provides immediate results. As we incorporate distributed rendering, our tool produces images of comparable quality without relying on expert-level 3D knowledge.

3 HOME STUDIO

Home Studio is a web-based tool implemented in Javascript that allows public Matterport scenes to be staged with furniture. Matterport's Bundle SDK acts as a connecting layer between our Three.js

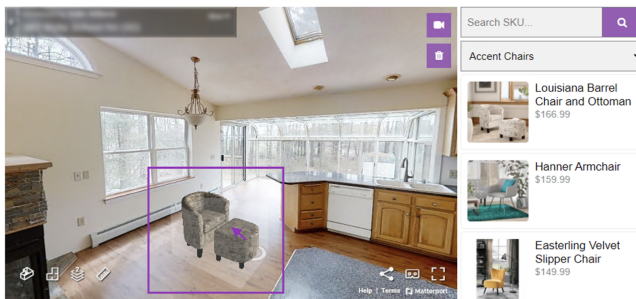


Figure 2: Home Studio's browsing interface, with an accent chair proxy model staged in the Matterport scene.

application and the Matterport showcase interface typically employed for virtual tours of real estate. Our tool enables photorealistic rendering of the staged layouts by reconstructing the Matterport view in an external rendering engine. In this section, we describe the different components of our system and how a user interacts with them to produce a final render.

3.1 Model Capture

Home Studio works as an add-on experience to a Matterport real estate model. This model can be either (1) a publicly available model, (2) a model scanned using the Matterport proprietary camera, or (3) a model scanned using the Matterport capture app on an iOS device. Creating the 3D model is a one time operation that involves moving throughout the space and taking panoramic shots until enough coverage of the desired staging space is captured. The Matterport capture app turns this collection of panoramic shots into a 3D model that can be used by Home Studio. No additional steps are required to make the model compatible with the Home Studio tool, making it retroactive to all existing Matterport scenes.

3.2 Virtual Staging

Home Studio offers a furniture catalog of tens of thousands of products by leveraging Wayfair's 3D Model API [32]. This includes a wide range of categories of both floor and wall mounted products with their 3D models optimized for real-time use. Home Studio allows users to browse products by category or search for specific products to include within the scene. The Matterport model viewer, which lets the user move through different panoramic shots of a property, is able to ingest external 3D models into the scene through the Bundle SDK. When a user selects a piece of furniture from the browsing interface, a 3D node is generated in the scene and placed directly in front of the user's viewpoint.

Clicking on any model within the scene lets the user change its location by panning their cursor on the floor or walls. Ray-casting to the scene's 3D model mesh is employed to determine where the object is re-positioned. The user can also change the orientation of the object by clicking and dragging the cursor over a model, making it face the direction of the cursor within the scene. Users can subsequently delete or re-position any 3D model before creating a render.

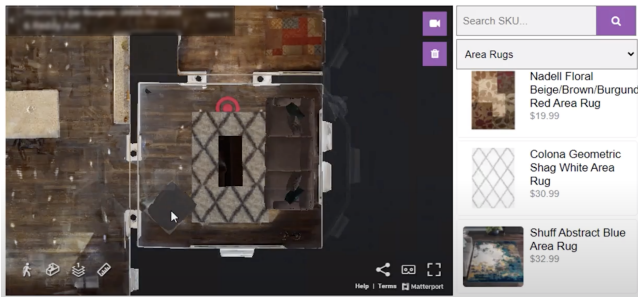


Figure 3: Designers are given a suite of tools for ease-of-use, including switching from a perspective to a top-down floor-plan view for a more holistic view of the designed layout.

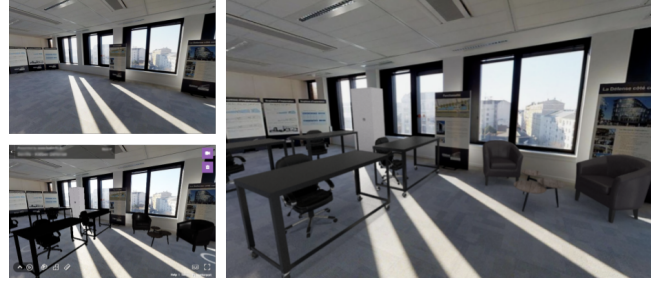


Figure 4: Unstaged publicly available Matterport scene from Redfin (top left); Low level of detail (LOD) 3D models staged (bottom left); Photorealistic render of the reconstructed, composited scene with high LOD models (right).

In addition to the aforementioned actions, users are able to perform standard Matterport virtual tour interactions, which include: re-positioning the camera to different panoramic shots within the model, panning the camera (akin to a 360 image viewer), and changing to an overhead floor plan view. While in floor plane mode, the user can still add, re-position, and rotate 3D models, giving them a more comprehensive view of the final layout.

3.3 Rendering

Once the user has staged products within the scene's 3D model, they can generate photorealistic renders from their viewpoint by clicking on the render button. This creates an HTTP request to a remote rendering service which includes: the transform of all products in view with respect to the camera, the camera's field of view, the high dynamic range (HDR) panorama of the current camera position within the scene, and the panorama's rotation with respect to the camera.

The payload is used to recreate a scene in 3DS Max on a remote server, which consists of a deconstruction of the more complex Matterport model into primary planes (floors & walls) generated from the normals of the products placed in the camera's view. These planes bounce light and shadows generated by applying the HDR panorama to the 3DS Max scene environment to relight the scene using image based lighting (IBL). A high resolution version of each 3D model is swapped in and re-positioned with respect to a V-Ray camera that is created to match the field of view of the Matterport scene's camera. This camera is then used to render the final product.

4 DISCUSSION

Home Studio satisfies two needs. Given Matterport captures are to-scale with an accuracy of 99% and Wayfair's 3D API models are designed to-scale, Home Studio is effectively size accurate. This allows users to see if a particular layout or product will fit in a space. Second, the high resolution renderings (albeit not real-time) offer a true to life visualization of how the product will in a space. Home Studio's simple drag and drop interface empowers non-experts, letting (1) renters/buyers visualize furnishings in a new home, (2) realtors create staged versions of a home to attract interest in a listing, and (3) design services offer realtors and home/business owners visualizations of project proposals.

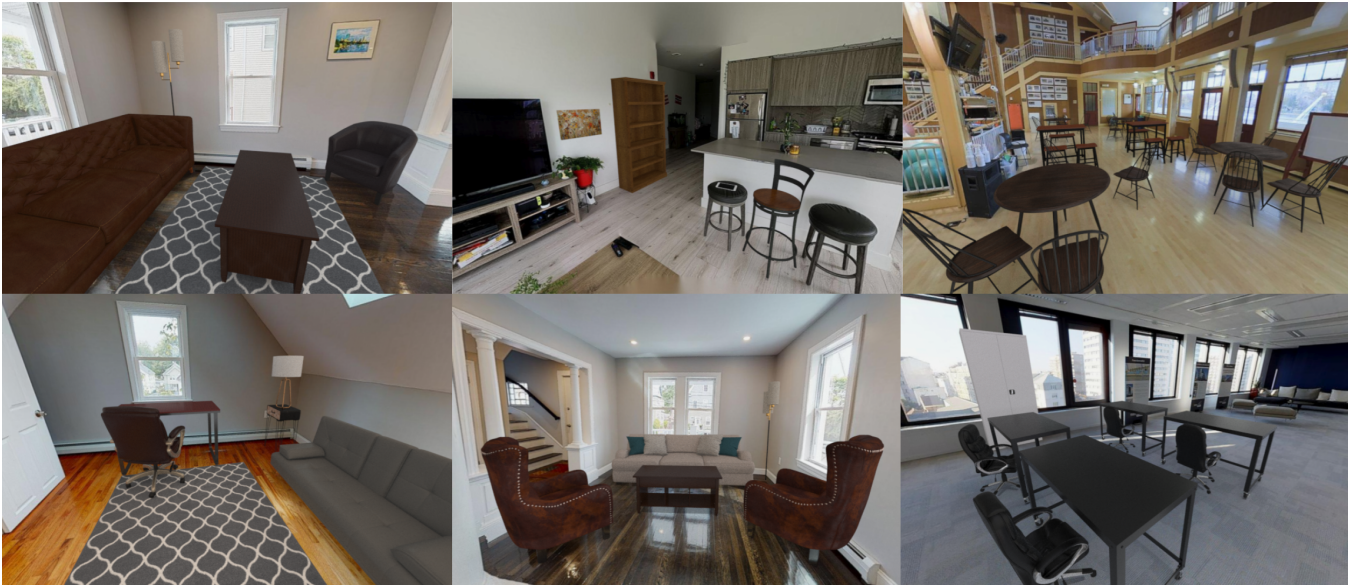


Figure 5: Six examples of photorealistic composites staged with high resolution models, using publicly available Matterport scenes from Redfin.

The quality of the output of our system stems from the swap-in technique employed in the rendering service. By using low level of detail (LOD) models in the real-time editor and swapping with high LOD equivalents during the reconstruction of the scene in 3DS Max, we enable a smooth experience without the need of dedicated rendering hardware on the client, while achieving higher quality than traditional AR experiences reliant on low LOD models. The use of high dynamic range (HDR) panoramic images to relight the scene further enhances the realism, translating the effect of light on the look of the staged products and contributing to the visual coherence of the composite. This effect is more pronounced in products with reflective materials like metal and glass.

Home Studio's integration with the Matterport SDK makes it usable on any publicly available space. As an experiment, we staged several properties listed on Redfin, and tested our tool with many other publicly available spaces (see Figure 5). This speaks to the ubiquity of our tool, which makes any previously captured scene actionable. We also captured a 2-bedroom apartment in roughly 10 minutes using the Matterport Capture app on iOS, which let us stage a home scanned with a smartphone with equal ease (see Figure 5 top-middle render).

Existing design services return proposals in a matter of days. Including the back and forth involved with the fine-tuning of a proposed design, most services real turnaround times for a particular project are in the order of weeks. Being driven directly from the 3D model and a HDR panorama, users don't need to worry about perspective matching, lighting, or shadowing. This empowers novice users, but also reduces the load on expert users, potentially reducing the time required to produce an appealing render. Back and forth between the designer and the client is supported by the tool, as opposed to verbal communication, which can reduce the time to convey ideas. Additionally, the Matterport virtual tour interface

lets customers request renders from their desired angles once a particular layout has been proposed by a designer, as opposed to being limited to the views offered in a typical design proposal.

Home Studio can also be employed as a virtual storefront for home goods. Retailers can leverage publicly available Matterport scenes to sell products in context or "sell the look." As opposed to limiting the retail experience to the traditional 2D website, a tool like Home Studio enables virtual brick-and-mortar experiences that mimic the way people shop, taking advantage of rich context, without incurring costs associated with real estate. Given our tool's tight integration with the Matterport viewer, its virtual reality view, which allows users to jump into the virtual tour using mixed reality headsets is supported by default, letting customers see their staged design in an immersive experience.

5 CONCLUSION

In this work, we have presented a novel web-based tool that enables a mixed reality staging experience on any publicly available Matterport scan. Home Studio allows users to create photorealistic room-scale visualizations of products in their space and at-scale. We have discussed how this tool can be employed by interior designers, business & home owners, realtors, and home goods companies to create meaningful mixed reality experiences that can instill confidence in the outcome of a project, enhance the way people shop with virtual storefronts, and explore new ideas for their homes. Our experience aims to lower the 3D skill bar required to generate high quality composites, enabling a drag and drop experience that can be leveraged by design experts and average homeowners alike [5].

6 FUTURE WORK

A continuation of this work requires us to evaluate the Home Studio tool with users to assess the usability of the system. We've recruited

a group of designers who will formally evaluate Home Studio, measuring its impact on the turnaround time and overall quality of visualizations submitted to their clients. We also intend to conduct a virtual survey to compare the visual quality of the renders generated by our system to other design services which use 3D artist labor to achieve similar results. We are also exploring tighter integration with other 3D scan formats, such as captures using Occipital's structure sensor, to enhance the ubiquity of the tool. Finally, we would like to incorporate real-time raytracing techniques to create a continuous feedback loop with designers and shorten the time to visualize designs.

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